

CSGA 2436: Realtime and Big Data Analytics

New York University Project Proposal

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Background

With the growing tourism industry worldwide, data is accumulating daily through hotel and restaurant reviews, social media updates, local public transportation logs, crime incident reports, and data from aviation and railway industries, etc. Various studies are utilizing big data to analyze tourism sentiments, convenience, and preferences for visiting different destinations. However, many of these studies focus on single-scope datasets, which may miss the nuanced trade-offs that tourists face.

For example, a hotel with excellent reviews might be predominantly visited by groups with more than one person, overlooking safety concerns and the low foot traffic in nearby metro stations. If a solo traveller books the hotel based solely on the favorable reviews, they could end up having less-than-ideal experience due to the limited traffic in nearby metro stations. Similarly, an area with high transportation accessibility and foot traffic may have restaurants with lower ratings on health inspection ratings.

This issue is particularly relevant in large metropolitan cities like New York City, where a wide variety of options and area dynamics come into play. Each of these factors contributes to the overall travel experience and the city's reputation as a global destination.

In this project, we propose an integrated analytics pipeline that uses the four main large-scale public datasets: hotel reviews (indicating traveler stay satisfaction and sentiment), NYC Restaurant Inspection Results (assessing food quality and cleanliness), MTA Subway Hourly Ridership (measuring transportation accessibility, hourly

foot traffic, which will serve as a proxy for perceived safety during non-rush hours), and NYPD Arrest Data (providing an indirect proxy for local safety and security).

By analyzing these datasets, the project aims to determine how several factors influence tourist satisfaction and attraction. The findings will help us understand the interplay between city dynamics and traveler decision-making. This could support (1) tourists in making informed decisions while visiting NYC, and (2) the development of data-driven urban design strategies to improve NYC as a tourist destination.

Project Overview

We will implement the workflow using big data technologies to ensure scalability and efficiency when handling large datasets. We will use MapReduce for data profiling and cleaning across all four datasets. After preprocessing, the cleaned data will be integrated and analyzed to understand the (co-)relationships between stay satisfaction, transportation accessibility, restaurant cleanliness, and neighborhood safety. We will use Apache Hive to create external tables on the integrated data and run SQL-like queries for correlation analysis and aggregations. For visualization and analysis, we will use Tableau to create interactive dashboards and Grafana to monitor trends and patterns dynamically.

Team Responsibilities

Each team member will be responsible for profiling and cleaning one dataset using MapReduce:

- **Aygun Najafova:** TripAdvisor Hotel Review Data
- **Era Sarda:** NYC Restaurant Inspection Dataset
- **Debdeep Naha:** MTA Ridership Data
- **Shivam Balikondwar:** NYC Crime Complaint Data

Design Diagrams

System Architecture

Figure 1 illustrates our overall system architecture. The four datasets are ingested into HDFS storage, where they undergo a three-stage processing pipeline (raw, cleaned, integrated) using MapReduce jobs. The integrated data is then queried using Apache Hive for correlation analysis and aggregations, with results visualized using Tableau and Grafana.

Diagram 1: System Architecture

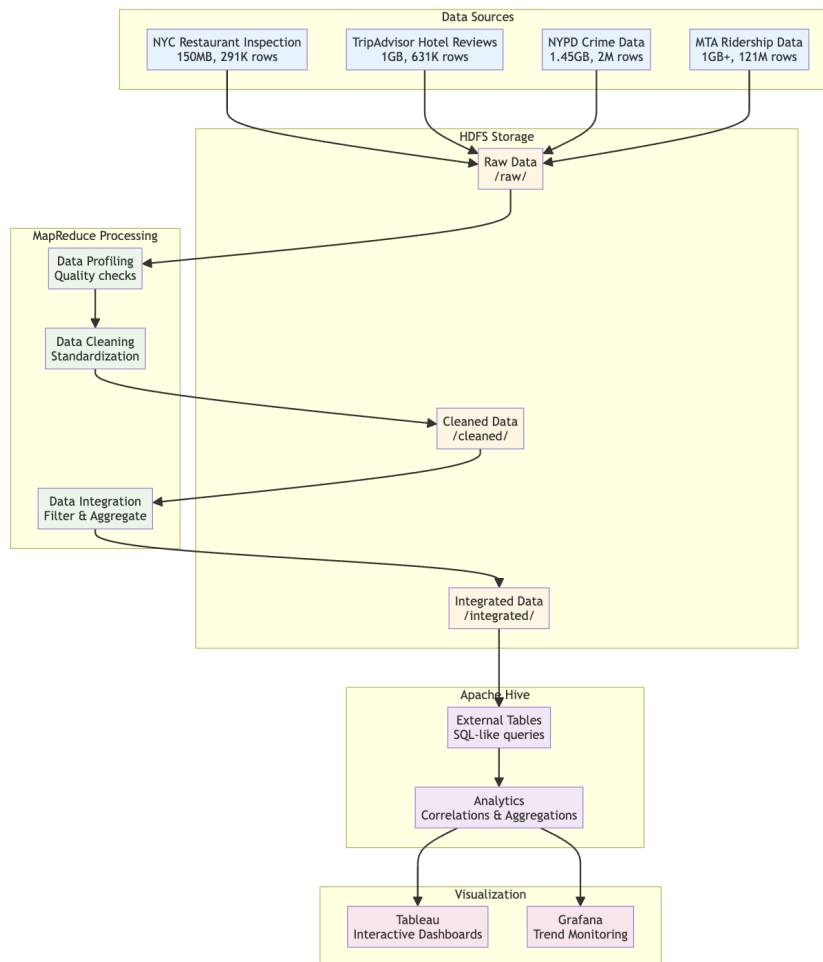


Figure 1: System Architecture showing data flow from sources through HDFS storage, MapReduce processing, Apache Hive querying, to visualization tools.

Data Integration Flow

Figure 2 shows how our four datasets are processed through MapReduce operations. The pipeline filters NYC entries, extracts analytics data from each column, and aggregates by location to create composite features.

Diagram 2: Data Flow and Integration

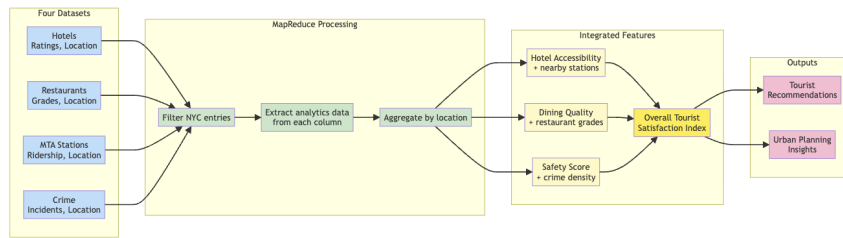


Figure 2: Data Integration Flow showing how MapReduce operations create integrated features for analysis.

About the Data

For our study we have primarily identified four datasets:

- **NYC Restaurant Inspection dataset [1]**
 - *Columns:* `camis` (unique restaurant ID), `dba`, `boro`, `building`, `street`, `zip-code`, `phone`, `cuisine_description`, `inspection_date`, `action`, `violation_code`, `violation_description`, `critical_flag`, `score`, `grade`, `grade_date`, `record_date`, `inspection_type`, `latitude`, `longitude`, `community_board`, `council_district`, `census_tract`, `bin`, `bbl`, `nta`, `location`.
 - *Data source:* Available from the City of New York Department of Health & Mental Hygiene (DOHMH) via the NYC Open Data portal.
 - *Size / Volume:* Approximately 150 MB, around 291,000 rows.
 - *Mode of use:* Downloadable from the portal, and no special approval needed.
- **TripAdvisor Hotel Review Data [4]**

- *Columns:* Review-text, Rating (hotel name/ID, review date, reviewer location).
- *Data source:* Shared publicly on Kaggle and scraped from TripAdvisor reviews.
- *Size / Volume:* Approximately 1 GB, about 631,300 rows.
- *Mode of use:* Downloadable through Kaggle.

- **NYC Crime Complaint Data [3]**

- *Columns:* 36 features including complaint number, date, time, borough, precinct, location (latitude/longitude), complaint type, descriptor, law category code.
- *Data source:* Provided by the City of New York Police Department (NYPD) via NYC Open Data.
- *Size / Volume:* Around 1.45 GB with over 2 million rows.
- *Mode of use:* We can download from the open data portal, and no special approval is needed.

- **MTA Ridership Data [2]**

- *Columns:* Has 12 columns mainly station_complex, transit_timestamp, fare_payment_class, ridership_count, borough, latitude, longitude, and other supporting fields.
- *Data source:* It is provided by the Metropolitan Transportation Authority (MTA) via the New York State Open Data portal.
- *Size / Volume:* It covers hourly ridership data across subway station complexes for years 2020-2024. It is in the above 1GB in size with 121 million rows.
- *Mode of use:* Downloadable from the Data.NY.gov portal, and no special approval is needed.

As an option we are also considering the Yelp API to scrape restaurant review data for NYC Restaurants.

If the hotel review dataset is insufficient, we will supplement our study by using the Yelp API to fetch restaurants in New York City. From this API, we will retrieve fields such as business name, categories, latitude, longitude, ZIP/postal code, star rating, review count and other available metadata. This backup dataset will complement our primary datasets not replace them and will help ensure we cover any gaps in hotel review data.

Notes on data access, ownership, size, and data characteristics.

- **Owner / Public or Private:** The city datasets (restaurant inspection, crime, MTA ridership) are owned by the City of New York (public open data). The hotel review dataset is publicly shared via Kaggle. The Yelp API data is provided by Yelp under developer terms.
- **Size / Volume:** The datasets range from hundreds of megabytes (e.g., 150 MB for NYC restaurant inspections) up to around a gigabyte (1 GB for hotel reviews).
- **Remarks:** All datasets will be treated as static snapshots for our study, we will download each once, store them in directories on our cluster, and not expect ongoing ingestion of new data.
- **How to access:** Downloadable via open data portals or third-party platforms (e.g., Kaggle); no special approval is needed for public datasets, though developer account/terms apply for APIs.
- **Additional remarks:** Each dataset contains individual records (e.g., one inspection record or one complaint or one ridership measurement) and includes location information (latitude/longitude, ZIP code, borough) which aligns with our geographic and analytical focus.

In summary, we will download and store each dataset in its own directory on our cluster, run MapReduce jobs on them, and treat them as static for reproducibility. The backup Yelp API dataset will likewise be a one-time extraction rather than a continuously updating job.

References

- [1] *DOHMH New York City Restaurant Inspection Results*. https://data.cityofnewyork.us/Health/DOHMH-New-York-City-Restaurant-Inspection-Results/43nn-pn8j/about_data. New York City Department of Health and Mental Hygiene.
- [2] *MTA Subway Hourly Ridership 2020–2024*. https://data.ny.gov/Transportation/MTA-Subway-Hourly-Ridership-2020-2024/wujg-7c2s/about_data. Metropolitan Transportation Authority.
- [3] *NYPD Complaint Data Current (Year To Date)*. https://data.cityofnewyork.us/Public-Safety/NYPD-Complaint-Data-Current-Year-To-Date-/5uac-w243/about_data. New York Police Department.
- [4] *TripAdvisor Hotel Review Data (via Kaggle)*. <https://www.kaggle.com/datasets/joebeachcapital/hotel-reviews/data>. TripAdvisor.